

Pranay Narsipuram

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EDUCATION

University of Trieste

Trieste, Italy

MSc in Data Science & Artificial Intelligence (EQF Level 7) | Weighted Avg: 24.75/30

Sep 2023 – Present

- Coursework: Probabilistic Machine Learning, Unsupervised Deep Learning, Advanced Programming (C++), MLOps, HPC & Cloud Computing, Ethics & Law of AI.
- Academic internship: 6-month industry placement at Realcomm S.R.L. embedded in the programme.

University of Rome Tor Vergata

Rome, Italy

BSc in Engineering Sciences – ICT & Internet Engineering (EQF Level 6) | Grade: 83/110

Sep 2018 – Oct 2022

- Thesis: *Soft Robotics in Medicine* (Supervisor: Prof. Fabrizio Quadrini).
- Research: LARM2 Robotics & Mechatronics Lab – bio-inspired robotic arm prototype with Prof. Marco Ceccarelli.
- Coursework: Physics, Mathematics, Embedded Systems, Laboratory of Sensors (Arduino, ESP32, Raspberry Pi).

EXPERIENCE

Application Software Developer

Nov 2024 – Apr 2025

Realcomm S.R.L. (IIoT & Industry 4.0)

Porcia, Italy – Hybrid

- Assigned to build a chatbot prototype as a microservice; initiated work with NER (Named Entity Recognition) to extract structured entities from unstructured text before integrating the full LLM pipeline.
- Developed NLP/NLU processing layers using SciSharp packages in C#; built on ASP.NET Core 8, Drupal 7, open-source LLMs, and an LLM orchestrator with a RAG architecture.
- Worked with Microsoft SQL Server for schema design, migrations, and data preparation; helped curate and structure reference data from Excel sheets for the retrieval pipeline – a notably tricky process given sparse and loosely organised source data.

Open-Source Software Developer Intern

Jun 2022 – Aug 2022

Invertase

Remote

- Worked on the Flutter Community `plus_plugins` monorepo under mentor guidance, supporting development of the `battery_plus` open-source plugin (Dart).
- Practiced open-source workflows: PR reviews, issue triage, and documentation improvements for a high-traffic Flutter ecosystem package.

PROJECTS

LCSA – Local Context Substrate for AI | *Rust, MCP, Cross-platform* | crates.io | [GitHub](https://github.com)

2025 – Present

- Built a layered signal model in Rust: Layer 1 raw platform events → Layer 2 structural signals (e.g. `ClipboardSignal`) → Layer 3 semantic signals (e.g. `code.updated`), giving AI agents change-aware context rather than snapshots.
- Implemented cross-platform clipboard, focus, and filesystem monitoring (Linux X11, macOS, Windows) with an explicit permission model that separates safe structural metadata from raw content access.
- Shipped `lcsa-mcp` as an MCP server exposing LCSA signals to AI tools (Claude Desktop, Cursor); published `lcsa-core` on crates.io.

Agent Governance Game | *AI Agents, Policy Design, Human-AI Interaction*

2025 – Present

- Building a simulation environment to study governance models for human-AI agent interactions, exploring policy constraints, incentive design, and trust calibration.

LARMbot Robotic Arm | *Robotics, Mechatronics, Soft Robotics* | LARM2 Lab, Univ. of Rome Tor Vergata

2020 – 2022

- Contributed to a bio-inspired robotic arm prototype drawing on musculoskeletal structural principles under Prof. Marco Ceccarelli; produced a bibliographic thesis on Soft Robotics in Medicine.

battery_plus Flutter Plugin | *Dart, Flutter, Open Source* | Invertase

2022

- Delivered cross-platform plugin contributions improving consistency across Android, iOS, macOS, and Linux.

TECHNICAL SKILLS

Applied (work & projects): LLMs, RAG, NLP/NLU, Named Entity Recognition, Microsoft SQL Server, ASP.NET Core 8, SciSharp, MCP (Model Context Protocol), Rust, Python, C#, C++, Dart/Flutter, SQL, Bash, Git/GitHub, Docker, Conda

Coursework & Academic: Probabilistic ML, Bayesian Inference, Unsupervised Learning, Autoencoders, Neural Networks, Backpropagation, PyTorch, Embeddings, Vector Databases, Anomaly Detection, HPC, Cloud Computing, MLOps

Frameworks & Tools: LLM Orchestrators, Open-source LLMs, Jupyter, IIoT / Industry 4.0

Systems / Embedded: Cross-platform systems programming (Linux, macOS, Windows), Arduino, Raspberry Pi, ESP32

Concepts: Agent Governance, Data Governance, VC Dimension, Bias-Variance Tradeoff, PAC Learning, Manifold Hypothesis

COMMUNITY, EVENTS & PUBLICATIONS

AI.dev: Open Source GenAI & ML Summit Europe 2025

Aug 2025, Amsterdam

- Attended summit covering open-source GenAI tooling, agentic frameworks, and ML infrastructure; exploring founding a local AI Tinkerers chapter in Trieste.

AI in Virology – AREA Science Park

Sep 2025, Trieste

- Seminar on Protein Language Models, protein–protein interaction modelling, and responsible AI training data for virology.

dev.to Article: “I Refused to Throw It Away” | dev.to/probis

2024

- Published technical/reflective piece on the dev.to community.

Volunteering: Associazione Liberi di Volare ONLUS

2024, Trieste

- Swift, swallow, and house martin conservation – fieldwork, informational sessions, and community outreach.